

Navier–Stokes 2D Advective Part

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1 General Formulation

We start from the conservative vectorial form of the Navier–Stokes equations (only the advective part):

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) = 0 \quad (1)$$

Let us isolate the time derivative:

$$\frac{\partial \mathbf{u}}{\partial t} = -\nabla \cdot (\mathbf{u}\mathbf{u}) \quad (2)$$

This equation is valid for any point in space. Since we are simulating in a 2D domain divided into small control volumes, we want to know how a cell and its variables change over time. Thus, we integrate equation (1) over an entire cell area:

$$\iint_{A_i} \frac{\partial \mathbf{u}}{\partial t} dA_i = \iint_{A_i} -\nabla \cdot (\mathbf{u}\mathbf{u}) dA_i \quad (3)$$

Assuming that the velocity field is uniform within the cell and applying the 2D Gauss divergence theorem:

$$A_i \frac{\partial \mathbf{u}}{\partial t} = \oint_{L_i} -\mathbf{u}\mathbf{u} \cdot \mathbf{n} dl_i \quad (4)$$

Distributing the unitary normal vector into each term:

$$A_i \frac{\partial \mathbf{u}}{\partial t} = \oint_{L_i} -\mathbf{u}(\mathbf{u} \cdot \mathbf{n}) dl_i \quad (5)$$

1.1 Cell Discretization

We discretize this equation over an unstructured triangular mesh. Let us define the enumeration convention for nodes and edges:

On the left-hand side, $\mathbf{u}_i$ denotes the cell-averaged velocity in cell i . On the right-hand side, the field variables are evaluated at each edge j . Replacing the contour integral with a sum over the three cell edges:

$$A_i \frac{\partial \mathbf{u}_i}{\partial t} = \sum_{j=1}^3 -\mathbf{u}_{ij} \mathbf{u}_{ij} \cdot \mathbf{n}_{ij} L_{ij} \quad (6)$$

Approximating the time derivative with a first-order forward Euler scheme where the superindex n denotes the time step:

$$A_i \frac{\mathbf{u}_i^{n+1} - \mathbf{u}_i^n}{\Delta t} = \sum_{j=1}^3 -\mathbf{u}_{ij}^n (\mathbf{u}_{ij}^n \cdot \mathbf{n}_{ij}) L_{ij} \quad (7)$$

Expressed in matrix form to solve for the next time step:

$$\begin{pmatrix} v_{xi} \\ v_{yi} \end{pmatrix}^{n+1} = \begin{pmatrix} v_{xi} \\ v_{yi} \end{pmatrix}^n - \frac{\Delta t}{A_i} \sum_{j=1}^3 \begin{pmatrix} (v_{xij}^n)^2 & v_{xij}^n v_{yij}^n \\ v_{yij}^n v_{xij}^n & (v_{yij}^n)^2 \end{pmatrix} \begin{pmatrix} n_{xij} \\ n_{yij} \end{pmatrix} L_{ij} \quad (8)$$

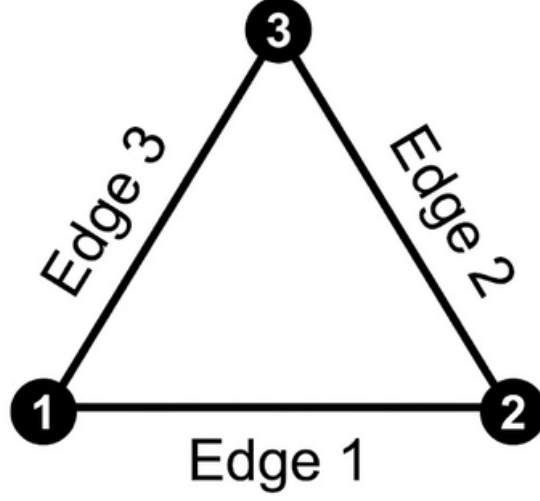


Figure 1: Triangular cell with counter-clockwise numbering of nodes and edges.

1.2 Interpolation Schemes On Edges

Since equation (8) requires the velocity field evaluated at the edges, but variables are stored at cell centroids, an interpolation scheme is required.

1.2.1 First Order Upwind

Let i be the target cell ID, k the neighboring cell ID, and j the edge shared between them. The average velocity components at the edge are defined as:

$$\begin{aligned} v_{x,\text{average}}^n &= \frac{v_{xi}^n + v_{xk}^n}{2} \\ v_{y,\text{average}}^n &= \frac{v_{yi}^n + v_{yk}^n}{2} \end{aligned} \tag{9}$$

The position vectors connecting the cell centroids to the edge j are given by:

$$\begin{aligned} \mathbf{n}_{ij} &= (n_{xij}, n_{yij}) \\ \mathbf{n}_{ik} &= (n_{xik}, n_{yik}) \end{aligned} \tag{10}$$

Since both vectors are exactly the same but in opposite direction, we can use just one. We have to calculate the inner products of the velocity average and our normal unitary vector, the result will tell us if the fluid is going from i to k or vice versa, the value that will give to the edge is the one from the source cell.

$$\text{If } \mathbf{v}_{\text{average}}^n \cdot \mathbf{n}_{ij} > 0 \quad \implies \quad \mathbf{v}_j^n = \mathbf{v}_i^n \tag{11}$$

$$\text{If } \mathbf{v}_{\text{average}}^n \cdot \mathbf{n}_{ij} \leq 0 \quad \implies \quad \mathbf{v}_j^n = \mathbf{v}_k^n \tag{12}$$

This interpolation scheme is so stable, easy and fast to add and use but adds a lot of numerical diffusion, which can modify our results very much.